

ENG507 Final Subjective

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2K25

What are Three dimensions of syllables?

The three dimensions of syllables are:

1. **Onset:** The consonant sound or cluster that begins the syllable.
2. **Nucleus** (or Peak): The vowel sound that forms the core or center of the syllable.
3. **Coda:** The consonant sound or cluster that ends the syllable.

What are Three plosive sounds?

The three plosive (or stop) sounds in English are:

1. /p/ (voiceless bilabial plosive)
2. /t/ (voiceless alveolar plosive)
3. /k/ (voiceless velar plosive)

What is Source-filter mechanism?

The source-filter mechanism is a model of speech production that explains how sounds are generated in the human vocal tract. It consists of two main components:

1. Source: The vocal cords (or glottis) produce a sound source, such as a buzz or a vibration, through vocal cord vibration.

2. Filter: The vocal tract, including the mouth and nose, acts as a filter that modifies the sound source, shaping its frequency characteristics and producing the desired speech sounds.

This mechanism allows for the production of a wide range of speech sounds, from vowels to consonants.

Voiced and voiceless sounds?

Voiced and voiceless sounds are classified based on the presence or absence of vocal cord vibration:

Voiced Sounds

1. Produced with vocal cord vibration
2. Vocal cords are close together and vibrate
3. Examples: /b/, /d/, /g/, /z/, /v/

Voiceless Sounds

1. Produced without vocal cord vibration
2. Vocal cords are apart and do not vibrate
3. Examples: /p/, /t/, /k/, /s/, /f/

Praat software?

Praat is a free, open-source software for speech analysis, created by Paul Boersma and David Weenink. It's widely used in phonetics, linguistics, and speech therapy.

Praat allows users to:

1. Record and edit speech sounds
2. Visualize and analyze acoustic properties (e.g., pitch, intensity, spectrograms)
3. Measure and annotate speech segments
4. Perform statistical analyses

Praat is available for Windows, macOS, and Linux.

Teacher resources for pedagogical development?

- Textbooks
- Online courses
- Workshops
- Conferences
- British Council Teaching English

Types of phonations?

There are mainly four possible glottis/larynx settings or types of phonations:

- ❖ **Voiceless** – when the folds are open apart and the air passing through the glottis freely (/t/ or /p/).
- ❖ **Voiced** – when the folds are tight together and there is vibration during the air passage through the glottis (e.g., /b/ or /d/).
- ❖ **Creaky voice** – when there is a slight opening in the front and the arytenoid cartilages are tight together, so that the vocal folds can vibrate only at the anterior end (the small opening at the top).
- ❖ **Breathy or murmuring sound** – when the vocal folds are apart but still, they are vibrating - a breathy voice is like a whisper except voice.

Three types of sounds related to approximants?

The three types of sounds related to approximants are:

- 1. Liquids**: /l/ and /r/ sounds, which are made by allowing air to flow freely through the mouth.
- 2. Glides**: /j/ and /w/ sounds, which are made by moving the tongue or lips towards a specific position.
- 3. Semivowels**: Sounds that are intermediate between vowels and consonants, such as /j/ and /w/.

Types of articulatory targets?

- 1) Place of articulation
- 2) Manner of articulation
- 3) Type of articulation

Spectrographic hints in Consents'?

1) Voiced

vertical striations corresponding to the vibrations of the vocal folds

2) Bilabial

Locus of both second and third formants comparatively low

3) Alveolar

locus of second formant about 1700–1800 Hz.

4) Velar

usually high locus of the second formant

5) Retroflex

general lowering of the third and fourth formants

6) Stop

gap in pattern (with burst for voiceless and sharp formant beginning for voiced stops)

7) Fricative

random noise pattern in higher frequency regions

8) Nasal

formant structure similar to that of vowels (with formants at 250, 2500, and 3250)

9) Lateral

formant structure similar to that of vowels (with formants at 250, 1200, and 2400)

10) Approximant

formant structure similar to that in vowels, usually changing

how to display recorded files?

To display recorded files, you can use various methods depending on the device, software, or platform used for recording. Here are some common ways:

For Audio Files

1. Audio Player Software: Use software like Audacity, Adobe Audition, or VLC Media Player to play and display audio files.
2. Waveform Display: Many audio editing software programs display audio files as waveforms, showing amplitude over time.

For Video Files

1. Video Player Software: Use software like VLC Media Player, Windows Media Player, or QuickTime Player to play and display video files.
2. Video Editing Software: Programs like Adobe Premiere Pro, Final Cut Pro, or DaVinci Resolve display video files and allow editing.

For Specific Platforms

1. Praat: For speech analysis, Praat software displays audio files as waveforms and spectrograms.
2. MATLAB: For scientific computing, MATLAB can display audio and video files using various toolboxes and functions.

what is the function of first formant in the vowels?

The first formant (F1) is a critical acoustic cue for vowel identification, providing information about vowel height, backness, and phonological contrasts.

what are the three sounds in the umbrella approximant?

The umbrella approximant, also known as the "w" sound, is actually a combination of three distinct sounds:

1. Bilabial Approximant: /w/ (as in "wet")

2. Labiovelar Approximant: /ʋ/ (a blend of lip rounding and velarization)

3. Velar Approximant: /ɰ/ (a soft, velar sound)

Together, these three sounds blend to form the characteristic "w" sound in many languages.

Glide and Their Formants?

Glides are also the sonorants (vowel-like) sounds as they have similar patterns (have formants). Read from our recorded file and take the first three formants (F1, F2 and F3) from the middle of the sounds for glides (both for /w/ and /j/) and explore their acoustic correlates. Carefully judge the center of these sounds (the midpoint of [w] and [j]). Analyze that how similar is the formant structure of glides with vowels and nasals. Draw lines to indicate F1, F2, F3 and compare with vowels. When you are done, try to answer the following questions:

1. Which vowel does each glide resemble?
2. Are there any differences between the contours, i.e. between the transitions out of the two glides?
3. What do you think may cause these differences (in terms of articulation)?

What is the function of second formant in vowels?

The second formant (F2) is essential for distinguishing vowel backness, rounding, and phonological contrasts, making it a critical acoustic cue for vowel identification.

Three Major suprasegmental features?

- 1) Pitch
- 2) Stress
- 3) Tone
- 4) Rhythm
- 5) intonation or juncture.

highlighted word of intonation?

The highlighted word related to intonation is **Pitch**

three types of pitch?

- 1) Rising Pitch
- 2) Falling Pitch
- 3) Level Pitch

Types of Articulatory Gestures? Explain each of one

Based on the definition of gesture as a possibility of the articulation of various speech sounds in contrast with English ones, the possible types of articulatory gestures are given here:

- 1) **Bilabial gesture** – very common in English (e.g., stops and nasal: p, b, m), In some languages (such as Ewe of West Africa), bilabial fricatives contrast with labiodental fricatives. The symbols for the **voiceless and voiced bilabial fricatives are [ɸβ]**. These

sounds are pronounced by bringing the two lips nearly together, so that there is only a slit between them. Ewe also contrasts voiceless bilabial and labiodental fricatives.

- 2) Labiodental fricatives – many languages (including English) have the **labiodental fricatives [f, v]**. But probably no language has labiodental stops or nasals except as allophones of the corresponding bilabial sounds. In English, **a labiodental nasal, [m]**, may occur when /m/ occurs before /f/, as in emphasis or symphony.
- 3)
- 4) **Dental sounds** are present both in British and American English, e.g. **dental fricatives [θ, ð]** but there are no dental stops, nasals, or laterals except allophonically realized (before [θ, ð] as in eighth, tenth, wealth). Many speakers of French, Italian, and other languages (such as Urdu, Pashto and Sindhi) typically have **dental stops such as [t̪ d̪]**. However, there is a great deal of individual variation in the pronunciation of these consonants in all these languages.
- 5) **Alveolar** are very common targets and stops, nasals, and fricatives all occur in English and in many other languages at alveolar as a target of articulatory gestures (e.g., t, d, n, l, r, etc.).
- 6) **Retroflex** is very common sound in many Pakistani languages which is made by curling the tip of the tongue up and back so that the tongue tip moves during the **retroflex sounds such as [ɳ, ɳ̌, ɳ̍, ɳ̎, ɳ̏]**. These sounds are also the feature of Indian English. **Palato-alveolar and palatal** are also possible articulatory gestures commonly found in world languages. Similarly, velar sounds found in Urdu and other Pakistani languages need to be mentioned here including [x, ɣ] which are velar fricatives. The gestures for pharyngeal (such as Arabic pharyngeal fricative [ʕ]) and epiglottal sounds (such as epiglottal fricative [ʕ̥]) involve pulling the root of the tongue or the epiglottis back toward the back wall of the pharynx. Many people cannot make a stop gesture at this position. But these sounds are found in Arabic and other Semitic languages.

Who gave the idea of Tube Models?

Actually, this is an old idea (initiated by the great German scientist Hermann Helmholtz) that a vowel is merely the rapid repetition of its peculiar note.

Creaky voice?

when there is a slight opening in the front and the arytenoid cartilages are tight together, so that the vocal folds can vibrate only at the anterior end (the small opening at the top).

Phonation Types?

In order to further explain the nature of the phonation as a process, we need to know the possible phonation types and its description. There are mainly four possible glottis/larynx settings or types of phonations:

- ❖ **Voiceless** – when the folds are open apart and the air passing through the **glottis freely (/t/ or /p/)**.
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- ❖ **Creaky voice** – when there is a slight opening in the front and the arytenoid cartilages are tight together, so that the vocal folds can vibrate only at the anterior end (the small opening at the top).
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What happen when aspirated stops are followed by a vowel?

when aspirated stops are followed by a vowel: voice onset time is greater than zero called a **positive VOT**. The length of the VOT in such cases is based on the practical measure of aspiration – the longer the VOT, the stronger the aspiration (Navajo, for example, has strongly aspirated stops where, the aspiration (and therefore the VOT) lasts twice as long as that of English - 150ms).

three dimensions of stress.

there are three dimensions:

1. **fixed word stress** (mainly found in Romance languages),
2. **variable word stress** (mainly found in languages such as English and German) and
3. **fixed phrase stress** (phrase as a third possibility as exhibited by Japanese) and they want to categorize languages on the basis of these three patterns